



K-ELECTRIC LIMITED

**BQPS-III 900MW CCP
HRSG INSTRUCTION
O&M Manual**

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Harbin Electrical International Company Limited

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1. FOREWARD

1.1 SUMMARIZE

The HRSG instruction is for Pakistan BQPS III Combined Cycle 900MW Power Plant Project.

This project shall install two sets of combined cycle generating units configured in 1-1-1 manner for single shaft arrangement, each set includes one gas turbine, one HRSG, one generator and one steam turbine. The general arrangement of HRSG is shown in drawing 129H-0.

The main features of HRSG are as follows:

HRSG is outdoor arrangement, without supplementary firing, natural circulation, and horizontal type, the two HRSG are in-line arrangement. Each HRSG has triple pressure (HP, IP and LP). Spray line is applied to control the temperature of SH & RH steam.

G is formed from inlet duct, heat exchange room, outlet duct and stack.

All heating surface of HRSG adopt flat serrated finned tubes, which are vertically arranged in heating exchange room. Upper and bottom headers are arranged at both ends of tubes. And two sling rods for each upper header are used to transfer the loads of tube and header to top beam.

Spacing between the heating surfaces are enough for maintenance, and manholes for repair are set on the side casing.

HP, IP and LP drums are arranged on top steel structure with drum supports.

HRSG is all-steel structure and self-supported the HRSG proper and auxiliary room are closed structure.

Inside insulation and inner casing are used for HRSG body.

HRSG operates in positive pressure, better seal and expansion structure are used for pipe penetrating through duct.

1.2 WARNING

The following safety instructions and measures help prevent operators from being injured during the operation and maintenance of this device. These safety regulations should not be considered the only precautions that should be taken. Operators should always use good judgment and take careful safety measures to perform related operations, and should not ignore the responsibility of ensuring the normal operation of the equipment based on the operation manual.

When operating equipment, safety is considered the primary principle, especially in the presence of pressurized gas, rotating equipment, high-voltage cables, flammable products and toxic substances. When individuals work in these situations, they are exposed to dangerous conditions. Unless you have been familiar with and have been trained to master the contents of this manual and accessories, you cannot try to operate this device and related equipment. In the process of performing related operations or working around the equipment, ensure that the protective equipment is worn correctly and that all relevant procedures and standards are always followed.

In order to prevent any hazards, the specified parts must be used when performing maintenance. For disassembly and assembly of parts, refer to relevant manuals and other factory materials to determine whether to use special tools.

Here is a warning: If you fail to follow the prescribed precautions, you will face the risk of serious injury, even death, and major property damage.

2. SAFETY

2.1 GENERAL SAFETY

Follow all safety regulations at the site where the equipment is operated and relevant operating procedures of the owner.

Place the nearest fire extinguisher in the operation and maintenance area of the device, and ensure that the path to the fire extinguisher is unobstructed for use in emergency situations.

Do not operate the equipment in poor visibility.

Do not clean or repair the equipment when the system is not depressurized.

Permission must be obtained when welding near the equipment.

Electrical equipment must be maintained by personnel with relevant qualifications.

Running parts must avoid running at high temperatures.

Keep the roads near the equipment and the equipment site channels completely unblocked.

The user should provide dynamic electricity and control power that meet the requirements, the raw gas that meets the design pressure, and the flanges, joints, pipes, valves, etc. which are outside of the skid and connected to the equipment supplied.

The safety valve and each safety protection setting value should not be changed without OEM recommendation. Safety valves, transmitters, instruments, etc. should be calibrated according to regulations.

Power, gas, and water should be cut off during maintenance, and an isolation blind flange should be installed as required, and warning signs should be hung on the corresponding parts of the equipment.

During use and storage, please make sure that the electric control box is protected from rain and snow, and no water leaks into the electric control box.

It is strictly forbidden to open the electrical panel door while energized! When discharging coolant and lubricant, the discharged liquid should be properly handled to prevent environmental pollution.

2.1.1 Personal Safety

Personal safety is “an individual's ability to go about their everyday life free from the threat or fear of psychological, emotional or physical harm from others.”

Personal safety in the workplace depends on your own awareness of potential threats and risks as well as your employer's safety policies and procedures.

The three basic rules for personal safety are:

1. Stay alert and be aware of your surroundings.
2. Give the impression that you are calm, confident and know where you are going.
3. Trust your instincts. If it doesn't look or feel right, it might not be.

2.1.2 Personal Protective Equipment

Personal protective equipment, commonly referred to as "PPE", is equipment worn to minimize exposure to hazards that cause serious workplace injuries and illnesses. These injuries and illnesses may result from contact with chemical, radiological, physical, electrical, mechanical, or other workplace hazards. Personal protective equipment may include items such as gloves, safety glasses and shoes, earplugs or muffs, hard hats, respirators, or coveralls, vests and full body suits.

Employers are also required to train each worker required to use personal protective equipment to know:

- When it is necessary
- What kind is necessary
- How to properly put it on, adjust, wear and take it off
- The limitations of the equipment
- Proper care, maintenance, useful life, and disposal of the equipment

If PPE is to be used, a PPE program should be implemented. This program should address the hazards present; the selection, maintenance, and use of PPE; the training of employees; and monitoring of the program to ensure its ongoing effectiveness.

2.1.3 Limb Protection

Using appropriate limb and body protection helps reduce the risk of injury when working with chainsaws and other equipment. Employers are responsible for providing limb and body protection for workers using chainsaws. This may include pants, chaps, or aprons made of cut-resistant material

2.1.4 Fall Protection

Fall arrest is a fall protection form that involves safely stopping a person from falling. Other forms of fall protection are fall guarding (protection preventing a person's access to a fall hazard area) and fall restraint (protection preventing the falling of persons working in a fall hazard area).

Depending on the purpose they serve, and the activities they are used for, OSHA recommends different types of fall safety equipment:

- Body belts restrain persons working in hazardous positions to reduce fall risk. They are for positioning only, so they cannot replace fall arrest systems.
- Chest harnesses can be used in environments with limited fall hazard that does not include vertical free fall, or for retrieval of persons from tanks, bins, or narrow spaces.
- Full body harnesses have the power to arrest the most dangerous falls.
- Suspension belts can be used to support workers needing to adjust their position vertically or to perform various activities while hanging.
- Rope lanyards are used as restraint means due to their elastic properties.
- Web lanyards are ideal for work environments with fall hazards under 2 feet.
- Cable lanyards work best in hot or corrosive environments but should be backed up by shock absorbing systems.
- Shock absorbers reduce fall arresting forces and fall injury risks.
- Rope grabs are deceleration devices travelling on lifelines, allowing safe ascension and descent of the worker or locking of the lifeline in case of fall.
- Retractable lifelines ensure fall protection and mobility in high fall hazard areas.
- Safety nets are best used for projects where there are no temporary floors or scaffolds, and the fall distance is higher than 25 feet.
- Rails can be used on fixed ladders and curved surfaces, to prevent falls.

2.1.5 Cleanliness

A clean workplace also ensures the health and safety of your employees and visitors while at the same time creating an environment that is pleasant to be in. The most important reason to maintain a clean job site is for the safety of all those that enter the site.

2.1.6 Safety on Site

Safety in maintenance is concerned with the freedom from danger, protection from or not being exposed to the risk of harm or injury during the process of

carrying out maintenance procedures. ... The results indicated that workers were aware of the safety practices during maintenance works.

8 workplace safety considerations every new business owner should consider

- Perform a job hazard analysis. ...
- Regular practice enforces safe habits. ...
- Enforce the rules of reporting. ...
- Provide personal protective equipment. ...
- Lead by example. ...
- Reward safety awareness. ...
- Think long-term. ...
- Ensure that breaks are taken.

2.1.7 Safety Checklist

Safety checklists are documents used during safety inspections for the identification of potential hazards. OSHA has provided a wide range of checklists for the identification of potential hazards in a variety of industries and applications

2.1.8 Electrical & Thermal Hazards

The grounding of electrical appliances and instrument control systems should be reliable, and the grounding resistance should not be greater than the design requirements.

Please keep the control box (cabinet, platform) clean and tidy, and do not place other objects, especially conductive objects, in the control box (cabinet, platform).

2.1.9 Mechanical Hazards

Mechanical hazards are hazards created by the use of or exposure to either powered or manually operated equipment, machinery or plant. Mechanical injuries are mostly caused either by contact or entanglement with machinery.

2.1.10 Noise & Vision Hazards

Exposure to prolonged or excessive noise has been shown to cause a range of health problems ranging from stress, poor concentration, and productivity losses in the workplace, and communication difficulties and fatigue from lack of sleep, to more serious issues such as cardiovascular disease, cognitive impairment, and tinnitus.

2.1.11 Pressure & Rupture Hazards

A most serious hazards in pressure vessels is the potential for cracking, which can lead to leaks, or to complete rupture, consequences of which include: Blast effects due to sudden expansion of vessel contents. Possible injuries/damage from fragmentation. Cracked and damaged vessels can result in leakage or rupture failures. Potential health and safety hazards of leaking vessels include poisonings, suffocations, fires, and explosion hazards. Rupture failures can be much more catastrophic and can cause considerable damage to life and property.

2.2 CAUTIONS SPECIFIC TO THIS SYSTEM

- a) This device takes the following measures to ensure safety:
- b) The solenoid valves and instruments used in this process are all explosion-proof products. Explosion-proof connectors are installed at the inlet and outlet of the power cable and signal cable.
- c) Multi-level safety protection systems such as safety valve, manual relief valve, PLC overpressure alarm, etc. are installed to ensure system safety ;
- d) Equipped with corresponding replacement interface, replacement valve, manual release pipeline, to provide conditions for nitrogen replacement ;
- e) With two-level protection function of alarm and shutdown ;
- f) Complete and fully automatic online monitoring system to ensure safety
- g) All control power circuits have insurance terminals to provide better electrical safety protection for the system as a whole.

3. System Overview

The main features of HRSG are as follows:

HRSG is outdoor arrangement, without supplementary firing, natural circulation, and horizontal type, the two HRSG are in-line arrangement. Each HRSG has triple pressure (HP, IP and LP). Spray line is applied to control the temperature of SH & RH steam.

G is formed from inlet duct, heat exchange room, outlet duct and stack.

3.1 HRSG'S BASIC FEATURES

3.1.1 HRSG'S BASIC PARAMETER

➤ **HRSG Type**

MHDB-SGT5-4000F-Q1

➤ **Type**

Triple pressure, reheat, no supplementary firing, horizontal type, natural circulation

➤ **Annual average pure condensing condition parameter**

HP main steam	Rated steam flow	313.08t/h
	HP steam outlet pressure	15.72MPa.a
	HP steam outlet temp.	567°C
IP main steam	Rated steam flow	50.56t/h
	IP steam outlet pressure	3.78MPa.a
	IP steam outlet temp.	278.3°C
RH steam	Rated steam flow	350.83t/h
	RH steam outlet pressure	3.52MPa.a
	RH steam outlet temp.	567°C
LP main steam	Rated steam flow	39.18t/h

	LP steam outlet pressure	0.45MPa.a
	LP steam outlet temp.	245.6°C
Feedwater temp.		55.8°C
Exhaust gas temp.		85.6°C
Gas resistance		≤3.3kPa (static pressure)
Heating surface area		399353 m ²
Total power consumption of auxiliary engine		~3217KW

3.1.2 THERMAL CALCULATION

➤ Summary sheet of thermal calculation

Case 1: 30T-70RH-100CC-ECoff-GAS

Gas flow : 2498.3 t/h

Item	Unit	HP SEC SH	HP RH	HP PRI-2 SH	HP PRI-2 RH	HP PRI RH	HP PRI SH	HP EVAP	HP SEC ECO	IP SH	IP EVAP	HP PRI3 ECO	LP SH	HP PRI2 ECO	IP ECO	HP PRI ECO	LP EVAP	LP ECO
Steam(water) flow	t/h	312.7	348.9	312.7	348.9	348.9	312.7	312.7	312.7	51.0	51.0	312.7	39.5	312.7	87.2	312.7	39.5	510.3
Steam(water)T in	°C	483.9	520.5	408.0	467.2	343.6	348.0	348.0	250.2	247.9	247.9	241.8	154.2	212.8	155.4	155.7	154.2	55.0
Steam(water)T out	°C	567.0	567.0	484.3	521.1	467.2	408.0	348.0	345.6	278.0	247.9	250.2	245.1	241.8	246.7	212.8	154.2	149.3
Steam(water)P out	MPa.g	15.60	3.39							3.65			0.35					
Gas T in	°C	616.9																
Gas T out	°C																	88.0

➤ Summary sheet of flus gas resistance calculation

Case 1: 30T-70RH-100CC-ECoff-GAS

Gas flow : 2498.32 t/h

Item	Unit	INLET DUCT	HP SEC SH	HP RH	HP PRI-2 SH	HP PRI-2 RH	HP PRI RH	HP PRI SH	HP EVAP	HP SEC ECO	IP SH	IP EVAP	HP PRI3 ECO	LP SH	HP PRI2 ECO	IP ECO	HP PRI ECO	LP EVAP	LP ECO	OUTLET DUCT	TOTAL (STATIC)
Gas Inlet Temp.	°C	—	616.90	592.20	579.60	551.10	536.50	502.00	458.10	354.00	293.20	291.30	259.10	254.60	251.70	251.70	223.60	195.10	160.80	—	—
Gas Outlet Temp.	°C	—	592.20	579.60	551.10	536.50	502.00	458.10	354.00	293.20	291.30	259.10	254.60	251.70	229.10	212.60	195.10	160.80	88.00	—	—
Gas Resis.	kPa	0.079	0.107	0.151	0.071	0.129	0.129	0.103	0.513	0.373	0.014	0.184	0.049	0.032	0.180	0.181	0.125	0.196	0.326	0.348	3.29
Gas Velo.	m/s	—	14.72	16.22	14.16	15.07	14.92	13.06	11.37	9.41	8.32	8.91	7.94	9.52	7.8	7.85	7.39	7.39	6.7	—	—

3.1.3 DESIGN CONDITION

➤ Location

This project is located in Malir district of Port Qasim, Karachi, Pakistan, 46 km from the capital of Sindh province Karachi city.

- **Temperature**
The temperature range to be considered for plant design is:
Minimum ambient temperature 0 °C
Maximum ambient temperature +50 °C
- **Rainfall**
The region where the plant locate is tropical monsoon climate, influenced by the monsoon, the maximum rainfall per day is 142.6mm.
- **Relative Humidity**
Maximum relative humidity (RH) 90%
- **Wind Direction and Velocity**
The dominant wind direction throughout the year is southwest wind (41%), the design wind speed is 113kph, and the wind pressure is 0.62kN/m².
- **Seismic**
Site Type: Class II
Customer-specified horizontal acceleration 0.24g
- **Fuel**
Natural Gas and light diesel are used as fuel for this Power Plant, (See the table below for fuel gas specifications).

Natural Gas Analysis Element	Volume %
CH ₄	84.7480
N ₂	3.1190
CO ₂	2.6590
C ₂ H ₆	8.1510
C ₃ H ₈	1.2160
nC ₄ H ₁₀	0.0530
iC ₄ H ₁₀	0.0400
C ₅ H ₁₂	0.0040
C ₅ H ₁₂ :2	0.0070
C ₆ H ₁₄	0.0030

Light Diesel Analysis Element	Volume %
C	85.9310
H ₂	13.4340
O ₂	0.0300
N ₂	0.0150
S	0.5900

➤ **Inlet flue gas data**

Gas-turbine operation condition	UNIT	1×1×1-100GT-30℃ NG	1×1×1-100GT-30℃ OIL
Exhaust Gas Flow	kg/s	694.361	687.933
Exhaust Gas temp.	℃	616.8	558.6
Gas components			
CO ₂	mol%	4.0632	4.6074
SO ₂	mol%	0	0.0118
N ₂	mol%	72.9413	75.6681
H ₂ O	mol%	10.2723	5.2496
O ₂	mol%	11.8687	13.5753
Ar	mol%	0.8544	11.8687

➤ **Demineralized water**

No.	Item	Unit	Data
1	(25℃) Electric conductivity	μS/cm	≤0.2
2	SiO ₂	μg/L	≤10
3	Water Hardness	μmol/L	~0

3.1.4 HRSG PERFORMANCE

➤ **HRSG thermal efficiency**

HRSG thermal efficiency reference HRSG Correction Curves.

➤ **Operation mode**

HRSG can be operated for constant pressure or sliding pressure by steam turbine.

The range of stable operation condition is 30%~100% condition.
Number of unit starts

Starting mode	Annual number of unit starts	Total number of starts during lifetime
Cold start(shutdown time>72h)	10	300
Warm start (10h<shutdown time≤72h)	260	7800
Hot start (shutdown time≤10h)	40	1200
In total	310	9300

➤ **Steam quality**

Steam quality should meet the standard of GB/T12145-2008 Quality criterion of water and steam for generating unit steam power equipment.
LP steam shall refer to HRSG operation instruction.

3.1.5 MAIN OUTLINE SIZE OF HRSG

Item	Unit	Data
Inlet Duct Center Line Elevation	mm	6500
Outlet Duct Center Line Elevation	mm	15075
Width of HRSG (outside of side casing)	mm	11660
Length of HRSG (Inlet Duct Fiber Expansion Joint to Stack CL)	mm	34620
Height of HRSG top casing	mm	28350
Elevation of HP Drum Center Line	mm	32350

Elevation of IP Drum Center Line	mm	32750
Elevation of LP Drum Center Line	mm	32500
Stack diameter	mm	7000
Stack Top Elevation	mm	60000

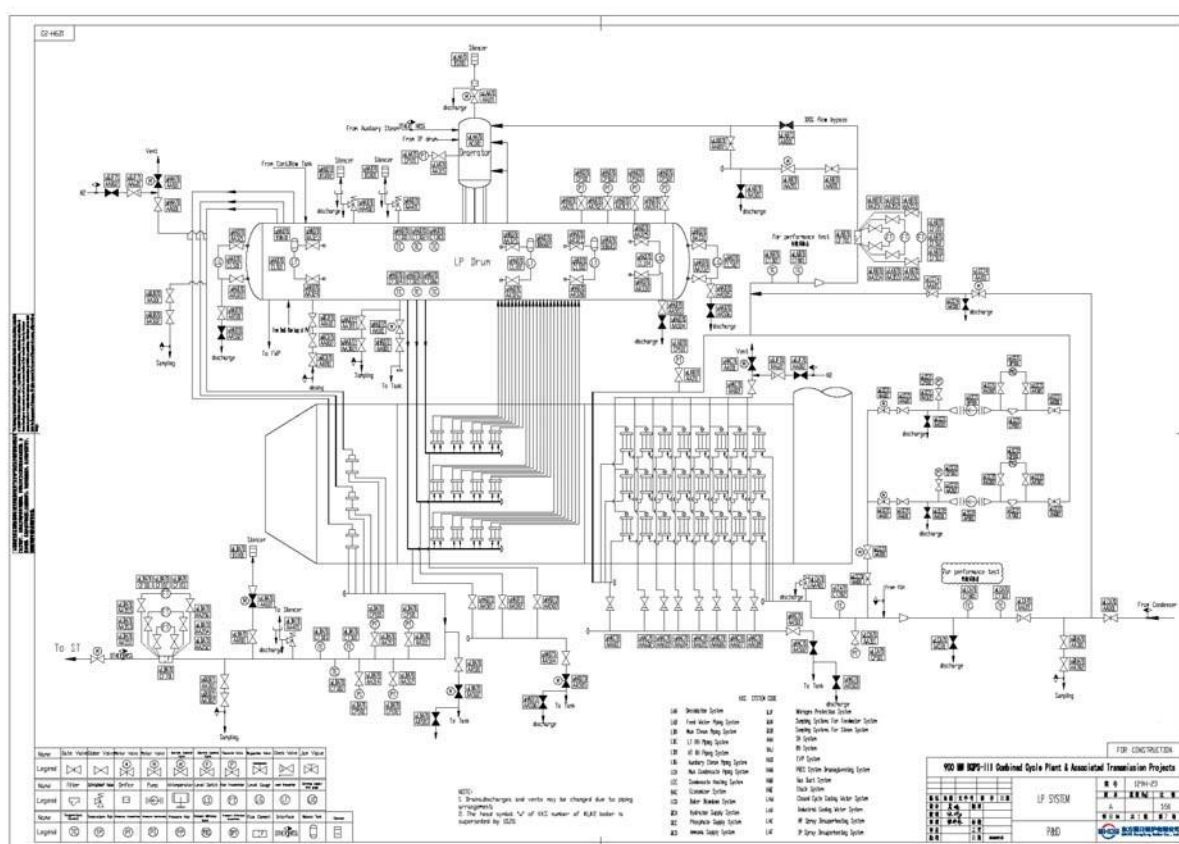
3.2 HRSG MAIN SYSTEM & BASIC STRUCTURE

3.2.1 STEAM & WATER FLOW PROCESS

HRSG steam & water system includes HP system, IP (RH) system, LP system (including deaerator system).

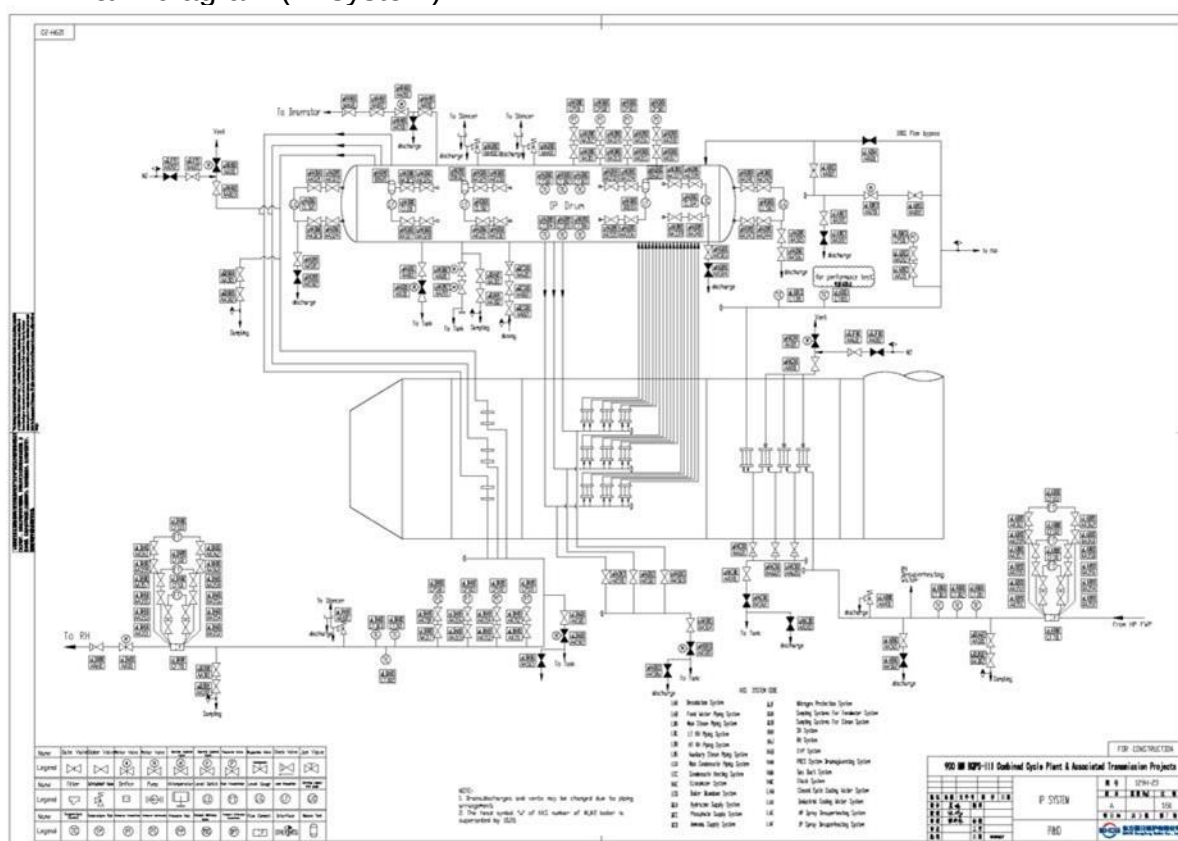
HRSG is equipped with deaerator, LP drum will be the deaerator water tank.

P&ID diagram (LP system):



LP system: condensate water (feed-water) flow through the LP ECON.,

P&ID diagram (IP system):



IP system: the water from high and medium pressure shared feed water pump extraction line flow through IP ECON, then flow to IP drum through control valve. Saturated water from IP drum is supplied to IP EVAP., water-steam mixture from evaporator's outlet will back to IP drum, and form the natural circulation. Saturated steam from IP drum flow through IP SH & cold

pump flow through HP feedwater control valve, HP PRI., SEC., TER. And FIN. ECON., then flow to HP drum. Saturated water from HP drum is sent to HP EVAP., water-steam mixture from evaporator's outlet will back to HP drum, and form the natural circulation. Saturated steam from HP drum flow through HP PRI., SEC. and TER. SH., and finally enter into HP cylinder of turbine. For controlling the HP SH steam temp., there are two spray

attemperators set up at SEC. and TER. SH. Connecting pipe and outlet pipe of SH.

The circulation ratio of each evaporator are as follows (performance guarantee condition)

Evaporator	HP	IP	LP
circulation ratio	5.9	38	51.8

HRSG hydrostatic test water capacity as follows:

Item	Unit	Data
HP drum	m ³	36.1
IP drum	m ³	22.2
LP drum	m ³	101.3
RH	m ³	33.6
HP SH	m ³	15.7
IP SH	m ³	2.7
LP SH	m ³	3.9
HP EVAP	m ³	36.5
IP EVAP	m ³	21.3
LP EVAP	m ³	28.4
HP ECON	m ³	68.2
IP ECON	m ³	6.8
LP ECON	m ³	55.3
CONN. piping	m ³	100
In total	m ³	532

3.2.2 GAS FLOW PROCESS

➤ Combined cycle

Gas from gas-turbine→ HRSG inlet duct→ HP TER. SH→ TER. RH→HP SEC. SH→ SEC. RH→PRI. RH→ HP PRI. SH →HP EVAP→HP FIN ECON→ IP SH→IP EVAP→ HP TER. ECON→LP SH→ IP ECON (HP SEC. ECON) → HP PRI ECON→ LP EVAP→ LP ECON→ outlet duct →stack→ atmosphere

3.2.3 BASIC STRUCTURE

- HRSG's heating surfaces are divided into 6 blocks along the direction of gas flow, and 3 blocks along the direction of HRSG's width, the details are as follows:

No.	Block's No.	Block's amount	Description of heating surface	Length (mm)	Width (mm)
1	Block I	3	HP TER. SH, TER. RH, HP SEC. SH,	23250	3612.5
2	Block II	3	SEC. RH, PRI. RH, HP PRI. SH	23250	3612.5
3	Block III	3	HP EVAP.	23550	3612.5
4	Block IV	3	HP FIN. ECON., IP SH, IP EVAP., HP SEC. ECON.	23550	3612.5
5	Block V	3	LP SH, IP ECON. (HP SEC. ECON), HP PRI ECON., LP EVAP.	23550	3612.5
6	Block VI	3	LP ECON.	23550	3612.5

Each block is consisted of several tube panels, which including upper header, bottom header and finned tube. The tube panels attachment, such as honeycomb support, anti-vibration baffle plate, anti-short gas passing plate, will be assembled into block within factory.

Honeycomb support: between upper and bottom header of each tube

panels, there are 14

sets of honeycomb supports in vertical direction of block I and block II, and 8 sets in block III, block IV, block V and block VI. It can effectively prevent the horizontal movement between tube and tube.

Anti-vibration baffle plate: each panel between upper and bottom headers, there is 1 set of anti-vibration baffle plate along the depth direction of HRSG, which can effectively prevent air-cell vibration.

Horizontal anti-short gas passing plate: there is arranged 1 set of horizontal anti-short gas passing plate on the part of tube which without fin of each panel.

Anti-seismic and supports device for headers: there is anti-vibration lug of upper and bottom headers, which can take the horizontal seismic force to steel structure by the tie bar.

Vertical anti-short gas passing plate: there is 1 set of vertical anti-short gas passing plate on both sides of each panel, which can prevent the formation of flue gas passage.

Bottom anti-short gas passing plate: there is 1 set of anti-short gas passing plate on the front and back sides of each block's bottom headers.

Upper anti-short gas passing plate: there is 1 set of anti-short gas passing plate on the front and back sides of each block's upper headers.

By arranging anti-short gas passing plate, it can effectively prevent the formation of flue gas passage, and avoid the influence on heat transfer effects.

Top hanger for panel: top hanger of each block is consisted of hanger rod, rocker and hanger frame. Top hanger is connected with the roof beam of HRSG, there is 2 hoist point of each upper header.

➤ **Drum**

The material of HP drum is SA-302MB, the size of HP drum is $\Phi 2070 \times 135$, and the length of shell is 13m. The heads of HP drum are spherical head. The

material of IP drum is SA-516GR70, the size of IP drum is $\Phi 1670 \times 35$, and the length of shell is 10m. The heads of HP drum are ellipsoidal head. The material of LP drum is SA-516GR70, the size of IP drum is $\Phi 3050 \times 25$, and the length of shell is 13m. The heads of HP drum are ellipsoidal head.

There are safety-valve, level gauge, **balance tank**, electrical contact level gauge, temp. element and other necessary components on each drum. The piping of continuous excretion (except LP drum) , add drug, feedwater and start-up drain is arranged on the drum along the Length Direction

There is the necessary nozzle for gauge on the drums. Drums are supported with sliding supports whose fixing points are at center line, drums can slide to two sides smoothly from center line.

Separators for steam and water are arranged in each drum. To ensure the steam quality, cyclone separator is set for first separation, the shutter and uniform distribution plate is for second separation for HP and IP drum, and underwater plate with orifices and uniform distribution plate is used for LP drum. Vortex eliminators are set at the inlet of downcomer to prevent from vortex.

➤ **HRSG's steel structure**

The steel structure and casing are combined as self-supported structure. The insulation and inner casing are arranged inside of casing. In height direction, the expansion center is at bottom of casing. The casing can expand upwards smoothly. In gas flow direction, the thermal expansion centre of HRSG is at the 5th column position. The thermal displacement of inlet duct and foreside casing is absorbed by non-metal expansion joint at inlet of HRSG. And that of rear casing is absorbed by non-metal expansion joint at outlet duct.

All finned tubes can expand downward smoothly.

➤ **Insulation and Inner Casing**

Insulation and inner casing are arranged inside of the inlet duct, HRSG's casing and outlet duct (before the non-metal expansion joint). Insulation material is paved staggered, and fixed with spring clamp and stainless steel

wire (or galvanized wire). This kind of insulation structure can not only meet the requirements when start-up and shut-down rapidly, but also reduce the noise of HRSG. Aluminum silicate fiber blanket is used in inner insulation, and the thickness of insulation can ensure good thermal insulation performance. The insulation's thickness of inlet duct is 280mm, HT area is 280mm and LT area is 120mm. Inner casing is paved as scale-type, which can effectively absorb thermal expansion. Inner casing uses SUH409L material stainless steel plate for all gas temp. area.

➤ **Inlet and outlet duct**

The inlet and outlet duct of HRSG consist of steel plate and type steel. The inlet duct connects with non-metal expansion joint directly, which can absorb thermal expansion of HRSG before the 5th column. There is arranged insulation and inner casing inside inlet duct. Non-metal expansion joint is also erected at outlet duct to absorb thermal expansion of HRSG after the 5th column. External insulation structure is used in outlet duct and stack.

➤ **Platform and Stairway**

The platform are almost arranged at left side of HRSG, or where the maintenance and operation is necessary for manhole, measuring point, valves and so on. The typical width of platform is 1 meter or wider if necessary. Platform will use the riffled plate.

3.2.4 STEAM TEMPERATURE CONTROL

➤ **Superheated steam temp. control**

For controlling superheated steam temp, there are arranged two stage spray type attemperator in SH system. Attemperators are separately settled in the connecting piping of HP SEC and TER SH and HP SH main steam pipe. The attemperators have multi-holes for spray, which are shown in the drawing 129H23.

➤ **Reheated steam temp. control**

For controlling reheated steam temp, there are arranged two stage spray type attemperators in RH system. Attemperators are separately settled in the connecting piping of SEC and TER RH and RH main steam pipe. The attemperators have multi-holes for spray, for which are shown in the

drawing 129H23.

3.2.5 PROTECTION FOR HRSG

➤ Protection for SH system

Protection for overpressure: safety valve is arranged on outlet pipe of HP, IP and LP SH to protect SH from over pressure. The setting pressure of safety valve is less than that of drum. The capacity of safety valve can assure the flow rate to cooling SH when safety valve pop up.

Temperature monitoring: thermal couples are arranged on inlet and outlet pipe of attemperator, outlet pipe of SH to monitor the temperature of SH.

➤ Protection for RH system

2 sets of safety valve are arranged on RH inlet piping, and 1 set is arranged on RH outlet piping. The setting pressure of outlet piping is less than that of inlet piping. The capacity of safety valve can assure the flow rate to RH when safety valve pop up. Thermal couples are arranged on inlet and outlet pipe of attemperator, outlet pipe of RH to monitor the temperature of RH.

➤ Protection for drum

2 sets of safety valves are arranged on each HP drum, IP drum and LP drum. Thermal couples are set on the top and the bottom of drum to monitor the wall temperature of drum. Inside metal temperature of steam side is also monitored by thermal couple. Through measuring and monitoring metal temperature of drum, control the temperature difference between upper and bottom, outside and inside wall within allowable scope to ensure the service life of drum.

In addition, local water level gauge and level transmitter are arranged on drums to monitor drum water level so that to ensure natural circulation safe and liable.

Visual level of drum is as follows:

Description	Level gauge		
	HP drum	IP drum	LP drum
Normal operation level (mm)	-50	-50	+400
(mm) Gauge visual length	>1150 (+525, -625)	>800 (+375, -425)	>2050 (+1025, -1025)

Center distance of sampling tube(mm)	1200 (+550, -650)	1000 (+500, -500)	2300 (+1150, -1150)
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Note: normal operation level, visual length and center distance of sampling tube are all based on geometric center line of drum

3.2.6 PIPING OF HRSG

➤ Steam piping of HRSG

Description	Size of pipe	Material
Pipe of HP main steam	φ273X28.58	SA-335P91
Pipe of hot RH steam	φ559×22.23	SA-335P91
Pipe of cold RH steam	φ559X22.23	SA-335P12
Pipe of LP main steam	φ406.4X9.53	SA-106B

➤ Feed water system

HP&IP system are equipped with 2×100% motor-driven constant speed BFP (1 for service, 1 for spare). There is arranged min. flow valve at the outlet of each BFP. Condensate system is equipped with 2×100% re-circulation pump (1 for service, 1 for spare). Re-circulation pumps are constant pump.

Water of condensate system is supplied to LP system. The water flow through LP ECON. and deaerator, then flow into LP drum.

For preventing low-temp. gas corrosion of LP ECON., there are arranged LP ECON. bypass line and re-circulation system. LP ECON. bypass line can decrease the volume of LP ECON. Feedwater and re-circulation system can improve the inlet water temp. of LP ECON. So that the low-temp. gas corrosion of LP ECON is prevented

IP feedwater is from LP drum. The water flow through IP ECON. by high and medium pressure shared feed water pump extraction line then flow into IP drum. Control valve is arranged at the outlet of IP ECON.

HP feedwater is from LP drum. The water flow through HP PRI, SEC, TER and FIN ECON. is sent to HP drum by high and medium pressure shared feed water pump. Control valve is arranged at the inlet of HP ECON., so that the flow of HP feed water can be adjust.

Date of High and Medium Pressure Shared Feed Water Pump & recirculation pump is as follows

Description	Unit	High And Medium Pressure Shared Feed Water Pump	Recirculation pump
Type		Horizontal, Constant speed	Constant speed, centrifugal
Amount	Set	2(1 for service, 1 for spare)	2(1 for service, 1 for spare)
inlet flow rate	t/h	347	250
Inlet pressure	MPa	0.63	1.26
Outlet pressure	MPa	17.55	2.3
Feedwater temp.	°C	155	148
Pumping head	MPa	16.92	1.04
Min. flow	t/h	130	70
Valid NPSH	mH ₂ O	21.9	107

➤ **Piping of attemperating water**

- a) SH attemperating water system: SH attemperating water is from HP feedwater. For controlling spray water flow, there is flow element and electric control valve, so that steam temp. can be controlled as well, and attemperating bypass is also set. There is a check valve after control valve, and a electric globe valve is set up at the main attemperating water line.
- b) RH attemperating water system: RH attemperating water is from IP feedwater. For controlling spray water flow, there is flow element and electric control valve, so that steam temp. can be controlled as well, and attemperating bypass is also set. There is a check valve after

control valve, and an electric globe valve is set up at the main attemperating water line.

➤ **HRSG blowdown system**

HRSG is equipped with intermittent & continuous blowdown tank. The design of blowdown system is taken full consideration of HRSG operation characteristic.

Continuous blowdown lines are set up for each drum. An electric control valve and an electric stop valve are arranged on HP & IP line. A motor stop valve and a manual stop valve are arranged on LP line.

Emergent blowdown lines (startup blowdown line) are set up for each drum. A motor stop valve and a manual stop valve are arranged on each line. Emergent blowdown lines connect to intermittent blowdown tank.

Intermittent blowdown line is set up at bottom of downcomer for HP, IP & LP evaporator. A manual valve and motor valve are arranged on each line. Intermittent blowdown lines are connected to intermittent blowdown tank.

➤ **Other piping system of HRSG**

- a) Drain line: drain lines are set up for each heating surface on lowest point of pipes. For operation during startup and shutdown, motor valve is equipped for the secondary drain valve so that it can be operated in DCS room.
- b) Sampling: sampling points are set up for condensate water, drum water, saturated steam, superheated steam.
- c) N2 line: N2 filling connections are set up for economizer, saturated steam connecting pipe, reheater inlet pipe for long time storage protection.
- d) Chemical dosing: chemical dosing connections of ammonia and hydrazine are set up at inlet of high and medium pressure shared feed water pump. HP and IP drums are equipped with connections of

hydrazine; LP drum is equipped with ammonia connections.

- e) Vent line: vent lines are set up on saturated steam line, superheater header, reheater header and HP/ IP/ LP economizer and highest points of pipe.
- f) Vent line: The safety valves are equipped with steam vent lines. Start-up vent lines are set up at outlet of HP and LP superheater. Silencers are set up in each steam vent line outlet to decrease the noise.

3.2.7 EXPANSION SYSTEM OF HRSG

Expansion center: stack takes bottom center as expansion center, and can upward expands freely. Proper of HRSG takes the cross point of the 5th column and center line of HRSG's bottom shell as expansion center. During operation condition, HRSG will expand in all directions freely from expansion center.

Height direction: header and heating surface tube take suspending level as expansion center, and downward expand. Internal equipment also expand without the limit of casing plate or anti-short gas passing plate.

Length direction: there is arranged expansion joint at the inlet & outlet duct.

Width direction: heating surface tube, upper headers and bottom headers take the center point of HRSG as expansion center, and expand along left-right direction.

There is restrict bracket at the carter of drum, so that the drum will expand along length direction of drum freely.

3.2.8 AUXILIARY EQUIPMENT OF HRSG

➤ **Vent silencer**

For reducing the noise, there are vent silencers at the outlet of each vent pipe.

➤ **Stack silencer**

For reducing the noise, there are stack silencers in the stack.

➤ **Stack**

Stack is steel structure, the diameter is 7m, and the elevation of outlet is 60m. There are arranged necessary platform and aviation warning light at the top

➤ **Arrangement of manhole**

For the facility of maintenance, there are manholes on the wall of inlet duct, outlet duct, stack and among the block.

4. Service and Support

4.1 Introduction

Mitsubishi Power Dongfang Boiler Co. Ltd.(MHDB), formerly known as Babcock-Hitachi Dongfang Boiler Co., Ltd. (BHDB), is a Sino-foreign joint venture company set up by Dongfang Boiler Group Co., Ltd. (DBC, 50% shares), Mitsubishi Power, Ltd. (45% shares) and Itochu Corporation (ITC, 5% shares) to market, design and manufacture boilers and environmental protection equipment, with a total investment of USD\$24 million, registered capital of USD\$22.2 million, and the registered place in No.1267, Chenggong Road, Jiaxing City, Zhejiang Province, China. As Babcock-Hitachi K.K. (BHK), one of the shareholders of former BHDB, was merged into Mitsubishi Hitachi Power Systems, Ltd., thus the company name of former BHDB was changed to that of MHDB.

Former BHDB was established on 1st July, 1996. The old plant located at No.1572, Sihaoqiao, Chengnan Road, covers an area of about 50000m² with the construction area of about 31000m². In order to meet the needs of the development, the company built new plant in Jiaxing Economic and Technological Development Zone, Phase I of which has been built and put into use, covering an area of about 69000m² with construction area of about 41000m². Now the new and old plants already have an annual output of 6 units of 9F class heat recovery steam generators (HRSGs), 20000 tons of coil heating surface, 20000 tons of H finned tube economizer, and 1800 tons of headers, 5000 tons of structural parts, with about 500 employees, and an annual output value of about 800 million yuan.

7.1 Contacting MHPS Dongfang

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7.2 Available Services

